

UNBROKEN CODES & BROKEN EMPIRES

A Historically Accurate Journey Through Cryptography and the Enigma Machine

Course: CS-4XX / ECE-4XX Modern Cryptography & Network Security

Focus: Classical Ciphers, Polish Cipher Bureau (1932), Bletchley Park, Shannon & Public Key Era

Era 1: Classical Antiquity & The Birth of Cryptanalysis

From Spartan Scytales to 9th-Century Frequency Analysis in Baghdad

1. Transposition & Monoalphabetic Ciphers

- **Scytale of Sparta (5th C. BC):** Strip of leather wrapped around a wooden rod. Earliest transposition cipher.
- **Caesar Shift Cipher (c. 58 BC):** Monoalphabetic substitution over Z_{26} :
 $C_i = (P_i + 3) \bmod 26$
- Used by Julius Caesar for military directives. Easily brute-forced since key space $|K| = 25$.

2. Al-Kindi & Frequency Analysis (c. 850 AD)

- **Al-Kindi (Baghdad):** Polymath at the House of Wisdom.
- Published *A Manuscript on Deciphering Cryptographic Messages*.
- **The Breakthrough:** Discovered natural languages contain skewed letter frequencies ($E \approx 12.7\%$, $T \approx 9.1\%$).
- **Impact:** Rendered all monoalphabetic substitution ciphers obsolete forever.

Era 2: Polyalphabetic Substitution & Breaking Vigenère

Defeating Single-Letter Frequency Analysis

1. The Vigenère Cipher (1553 / 1586)

- Formulated by Giovan Battista Bellaso, popularized by Blaise de Vigenère.
- Uses a repeating keyword K to shift letters independently:
$$C_i = (P_i + K_{\{i \bmod m\}}) \bmod 26$$
- Flattens single-letter frequency distribution. Dubbed '*Le Chiffre Indéchiffrable*' for 300 years.

2. Cracking the 'Unbreakable' Cipher (1854 / 1863)

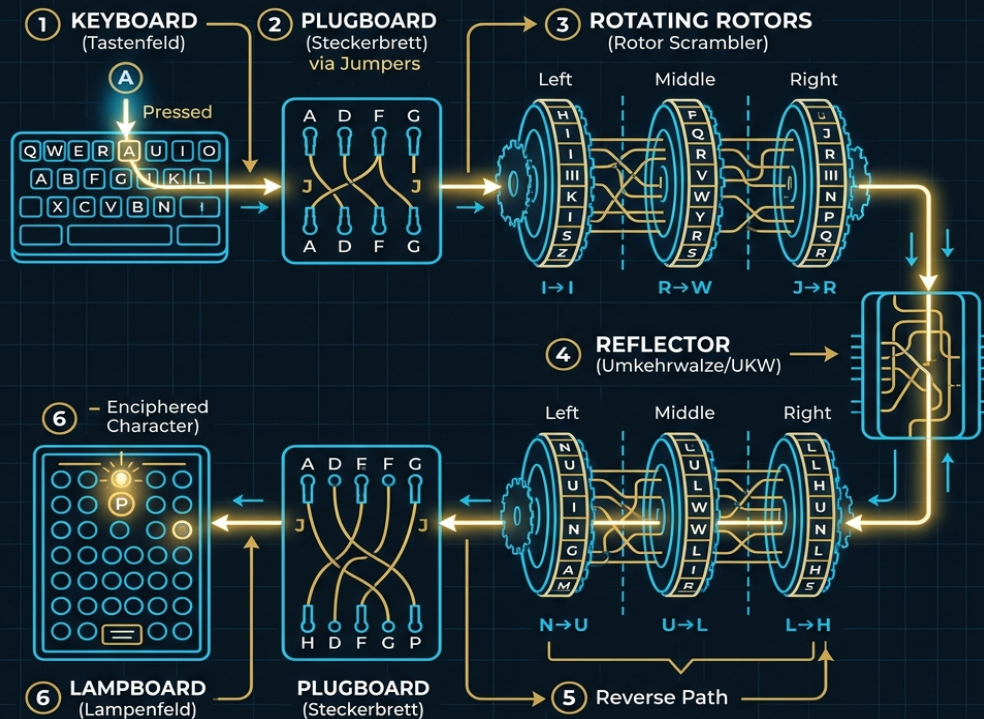
- **Charles Babbage (1854) & Friedrich Kasiski (1863)**: Independently cracked Vigenère.
- **Kasiski Examination**: Distance between repeating n -grams reveals key length m .
- **Index of Coincidence (IC)**: Developed by W.F. Friedman:
$$IC(T) = \sum f_i(f_i - 1) / [N(N - 1)]$$

English text $IC \approx 0.0667$ vs Random text $IC \approx 0.0385$.

Era 3: The Mechanics of the WWII Enigma Machine

Arthur Scherbius Patent (1918) & German Military Adoption

MECHANICAL ENIGMA MACHINE: ELECTRICAL SIGNAL FLOW (WWII)



Enigma Hardware & Signal Path

1. **Keyboard**: Keypress sends electrical current.
2. **Plugboard (Steckerbrett)**: Swapped 10 pairs of letters before & after rotors.
3. **Rotors (Walzen)**: 3 interchangeable rotors chosen from 5 (Kriegsmarine used 4/8). Rightmost rotor stepped every keypress.
4. **Reflector (Umkehrwalze)**: Bounced signal back through rotors in reverse.
5. **Lampboard**: Final encrypted letter lit up.

Astronomical Key Space:

~ **158,962,555,217,826,360,000 Key Combinations!**

The Unsung Heroes: Polish Cipher Bureau (Biuro Szyfrów)

The Mathematical Breakthrough 7 Years Before Bletchley Park (1932–1939)

1. Marian Rejewski's Mathematical Proof (1932)

- 27-year-old Polish mathematician at Biuro Szyfrów in Warsaw.
- Applied **Permutation Group Theory** to double-encrypted message indicators.
- Derived internal wiring equations of all 3 German rotors **without ever seeing a military Enigma machine!**

2. Electromechanical Hardware & Pyry Handover (1938-1939)

- **The Bomba (1938):** Rejewski built electromechanical units using 6 synchronized Enigmas to search for daily rotor orders.
- **Zygalski Sheets:** Perforated paper sheets created by Henryk Zygalski to track rotor positions.
- **July 1939 Pyry Handover:** Realizing German invasion was imminent, Poland handed over replica Enigma machines, Bomba designs, and math proofs to Britain and France.

Bletchley Park, Alan Turing & The Ultra Secret (1939–1945)

Exploiting Enigma's Fatal Architectural Flaw

1. Enigma's Fatal Design Flaw

- Enigma contained a fatal electrical flaw:
A letter could NEVER encrypt to itself! ($E(x) \neq x$)
- **Cribs:** Predictable text in military cables (e.g. weather reports 'WETTERVORHERSAGE').
- If a crib matched a ciphertext letter at the same index, that alignment was mathematically impossible and ruled out instantly!

2. The Turing-Welchman Bombe & ULTRA Impact

- **Alan Turing (Hut 8) & Gordon Welchman:** Designed the electromechanical **Turing Bombe**.
- Used Welchman's *Diagonal Board* to exploit plugboard symmetries and test thousands of rotor settings per second.
- **Historical Impact:** ULTRA intelligence shortened WWII in Europe by **2 to 4 years**, saving millions of lives.

Era 4: Claude Shannon & Symmetric Block Ciphers

Information Theory, Confusion, Diffusion & AES-256

1. Claude Shannon's Information Theory (1949)

- Published *Communication Theory of Secrecy Systems*.
- Formulated **Confusion** (obscuring key-ciphertext link) and **Diffusion** (spreading plaintext redundancy).
- Proved **Perfect Secrecy** for One-Time Pad: $P(M=m \mid C=c) = P(M=m)$.

2. DES to AES-256 Standardization

- **DES (1977)**: 56-bit Feistel Network block cipher. Broken by brute force in 1998.
- **AES (2001)**: Substitution-Permutation Network over Galois Field $GF(2^8)$ matrix operations.
- Standardized with 128, 192, and 256-bit key sizes. Powers modern disk and TLS data encryption.

Era 5: The Public-Key Cryptography Revolution

Solving the Key Distribution Paradox with Asymmetric Math

1. Diffie-Hellman & RSA (1976-1977)

- **Diffie-Hellman (1976):** Whitfield Diffie & Martin Hellman solved open key exchange using Discrete Logarithm Problem ($g^x \bmod p$).
- **RSA (1977):** Rivest, Shamir, & Adleman introduced trapdoor one-way functions based on Prime Factorization ($N = p \cdot q$).

2. Elliptic Curve Cryptography (ECC)

- Koblitz & Miller (1985) introduced Weierstrass curves:
$$y^2 = x^3 + ax + b \bmod p$$
- **ECC Efficiency:** 256-bit ECC (Curve25519) matches 3072-bit RSA security level.
- Powers modern HTTPS web traffic, SSH keys, Signal messaging, and Bitcoin transactions.

Era 6: The Quantum Threat & Post-Quantum Cryptography (PQC)

Preparing Global Security for Shor's Algorithm

1. Shor's Quantum Threat (1994)

- Running on a Cryptographically Relevant Quantum Computer (CRQC), Shor's Algorithm solves prime factoring and discrete logs in **polynomial quantum time $O(n^3)$** .
- **Result:** Quantum computers will break RSA, DHKE, and ECC simultaneously.

2. NIST Post-Quantum Standards (2024)

- NIST finalized Post-Quantum Standards based on **Lattice Mathematics**:
- **ML-KEM (CRYSTALS-Kyber)**: PQC Key Encapsulation Mechanism.
- **ML-DSA (CRYSTALS-Dilithium)**: PQC Digital Signature Standard.
- **SLH-DSA (SPHINCS+)**: Hash-based stateless signature standard.

Summary: Core Laws of Security & Cryptography

Lessons for Computer Engineers & Cybersecurity Professionals

4 Immutable Engineering Principles of Cryptography

1. **Kerckhoffs's Principle:** System security must depend solely on key secrecy, NEVER on keeping the algorithm hidden ('Security by Obscurity' fails).
2. **Mathematics > Hardware:** A single architectural flaw (e.g. Enigma's $E(x) \neq x$) invalidates billions of key combinations.
3. **Never Reuse Nonces or Key Material:** OTP key reuse (Venona) & AES-GCM nonce reuse reduce stream ciphers to plaintext XOR.
4. **Don't Roll Your Own Crypto:** Always use audited standard implementations (e.g. libsodium, OpenSSL).